Core CS

DBMS — Complete Interview Notes

Database fundamentals for interviews: ER diagrams, normalization, ACID, transactions, indexing, concurrency control, and NoSQL vs SQL.

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Database: A database is a collection of related data which represents some aspect of the real world. A database system is designed to be built and populated with data for a certain task

Database Management System (DBMS) is a software for storing and retrieving users' data while considering appropriate security measures. It consists of a group of programs which manipulate the database. The DBMS accepts the request for data from an application and instructs the operating system to provide the specific data. In large systems, a DBMS helps users and other third-party software to store and retrieve data.

Database management systems were developed to handle the following difficulties of typical File-processing systems supported by conventional operating systems.

1. Data redundancy and inconsistency
2. Difficulty in accessing data
3. Data isolation – multiple files and formats
4. Integrity problems
5. Atomicity of updates
6. Concurrent access by multiple users
7. Security problems

1. Data redundancy and inconsistency
Same data duplicated in multiple files. Updates in one place don't sync everywhere → inconsistent values (e.g., same customer has different addresses in billing vs. shipping files).

2. Difficulty in accessing data
No flexible query language. To get new reports, you must write a new program in a low-level language (e.g., COBOL) – cannot just ask "get all customers above 30".

3. Data isolation – multiple files and formats
Data scattered across different files with different structures (.txt, .dat). Combining them requires complex, custom code.

4. Integrity problems
No automatic rules to enforce valid data (e.g., age > 0, salary range). Must be coded manually in every program – easy to forget.

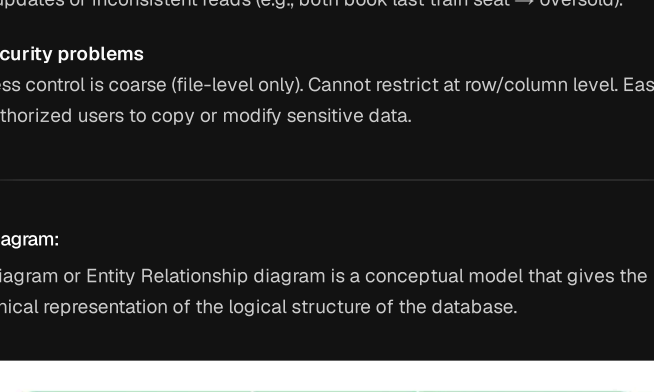
5. Atomicity of updates
If system crashes during a multi-step update (e.g., transferring ₹1000: debit account A, credit account B), only one step completes → money disappears or is created. No "all-or-nothing" guarantee.

6. Concurrent access by multiple users
No built-in locking. Two users reading and writing same file simultaneously causes lost updates or inconsistent reads (e.g., both book last train seat → oversold).

7. Security problems
Access control is coarse (file-level only). Cannot restrict at row/column level. Easy for unauthorized users to copy or modify sensitive data.

ER diagram:

ER diagram or Entity Relationship diagram is a conceptual model that gives the graphical representation of the logical structure of the database.



Figures	Symbols	Represents
Rectangle	▭	Entities in ER Model
Ellipse	⬭	Attributes in ER Model
Diamond	◇	Relationships among Entities
Line	—	Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse	◎	Multi-Valued Attributes
Double Rectangle	▣	Weak Entity

1. Entity
Real-world object (noun) – e.g., *Student, Car, Order.*
→ **Strong:** exists independently.
→ **Weak:** depends on another entity (needs foreign key).

2. Attribute
Property of an entity – e.g., *Name, Age, Price.*
Types:

- **Simple** (atomic) – e.g., Age
- **Composite** – e.g., Full Name (First + Last)
- **Derived** – e.g., Age (from DOB)
- **Multivalued** – e.g., Phone Numbers (multiple)

3. Key Attribute
Uniquely identifies an entity – e.g., *Student ID.* (Underlined in diagram).

4. Relationship
Association between two or more entities – e.g., *Enrolls* (Student ↔ Course).
→ Shown as a **diamond**.

5. Degree of Relationship
Number of entities involved:

- **Unary** (1 entity) – e.g., Manager manages Employee
- **Binary** (2) – most common
- **Ternary** (3+) – rare

6. Cardinality Constraints (Mapping)
How many entities participate:

- **1:1** (one-to-one)
- **1:N** (one-to-many)
- **M:N** (many-to-many)

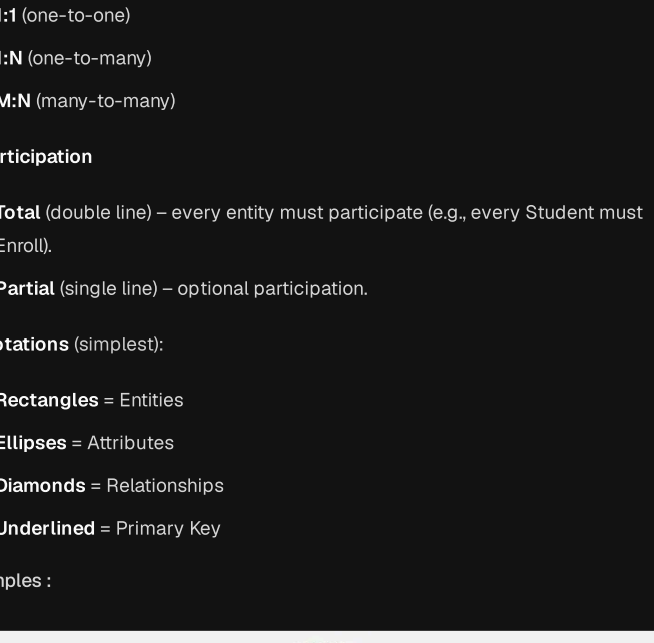
7. Participation

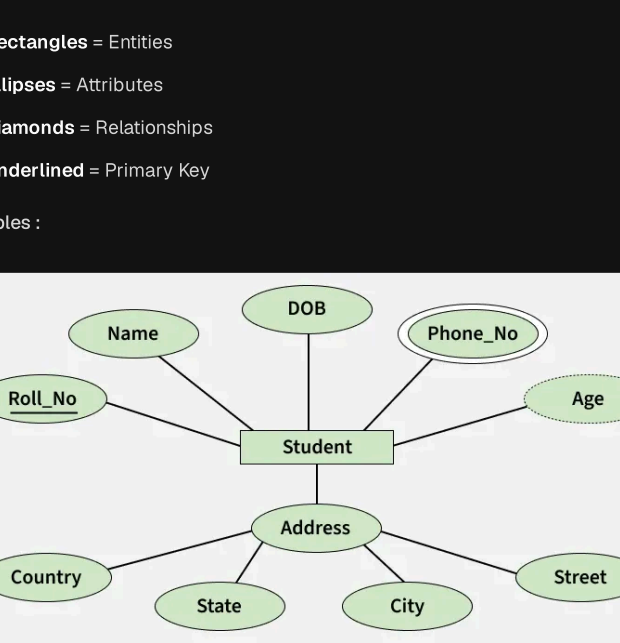
- **Total** (double line) – every entity must participate (e.g., every Student must Enroll).
- **Partial** (single line) – optional participation.

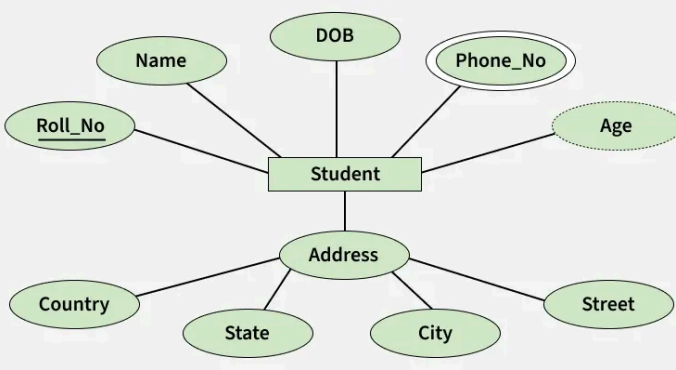
8. Notations (simplest):

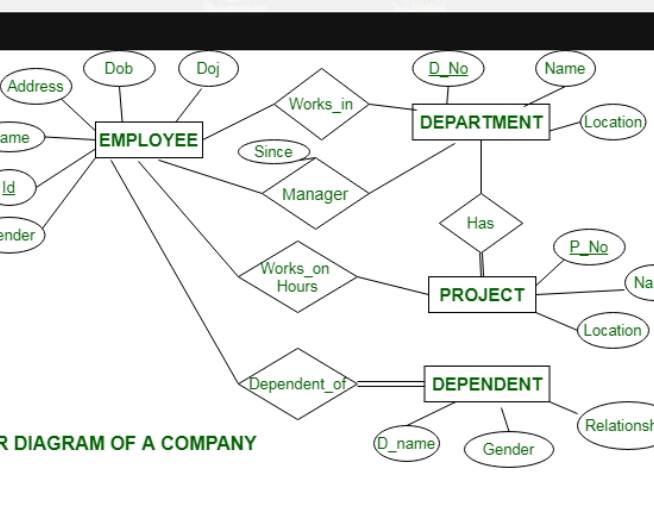
- **Rectangles** = Entities
- **Ellipses** = Attributes
- **Diamonds** = Relationships
- **Underlined** = Primary Key

Examples :









Payment is Weak Entity bcz it depends on the Officer. Payment number is Discriminator(**partial key**). Discriminator can identify a group of entities from the entity set.

- **Strong entity:** `Course` (PK: `Course_ID`)
- **Weak entity:** `Section` (depends on Course)
- **Discriminator:** `Sec_No` (1, 2, 3...) → `Section` is uniquely identified by (`Course_ID` + `Sec_No`)
  → `Sec_No` alone is not unique across all courses, but it is unique *within* one course.

Relationship = Association among entities.

Degrees (by number of participating entity sets):

Type	Entities Involved	Example
Unary (Recursive)	1	*Employee* **manages** *Employee* (boss-subordinate)
Binary	2	*Student* **enrolls in** *Course* (most common)
Ternary	3	*Doctor* **prescribes** *Medicine* to *Patient*
N-ary	N (any number)	*Supplier* supplies *Parts* to *Project* (4 entities)

Type	A → B	B → A	Example
1:1	At most 1	At most 1	*Person ↔ Passport*
1:N (One-to-Many)	Any number (0+)	At most 1	*Department → Employee* (one de
N:1 (Many-to-One)	At most 1	Any number (0+)	*Employee → Department* (many
M:N (Many-to-Many)	Any number (0+)	Any number (0+)	*Student ↔ Course*

Attributes :

Type	Definition	Example
Simple	Cannot be divided further	Age, Salary
Composite	Made of smaller simple attributes	Name (First + Middle + Las
Multivalued	Can have multiple values for one entity	Mobile No (home, work), En
Derived	Derived from other attributes	Age (from DOB), Experienc
Key	Uniquely identifies an entity	Roll No, Student ID, PAN
Relational constraints :		

Constraint	What it ensures
Domain	Attribute values must be from a defined set (atomic)
Tuple Uniqueness	No two rows (tuples) are identical in a relation
Key	Primary key values must be **unique** and **not null**
Entity Integrity	**No part** of the primary key can be NULL
Referential Integrity	When a table has a **Foreign Key (FK)**, that FK must point to a **valid, exist**

Key	Definition	Key Points
Super Key	Any set of attributes that uniquely identifies a tuple	Can have extra/unnece
Candidate Key	Minimal super key (no extra attributes)	A super key reduced to
Primary Key	One candidate key chosen by the designer	**Unique + NOT NULL**; on
Alternate Key	Candidate keys **not** chosen as primary key	Also called secondary k
Foreign Key	Attribute that references PK of another table	Values must exist in par
Composite Key	Primary key made of **two or more** attributes	Use when single attribu
Unique Key	Uniquely identifies each record	**Unique** but **can be NUL**

	Primary Key	Unique Key
Uniqueness	✅ Yes	✅ Yes
NULL allowed	❌ No	✅ Yes (one NULL allowed)
Can be updated	✅ Yes (but risky)	❌ No (non-updatable)
Count per table	Only 1	Multiple allowed

Functional Dependencies

Imagine one single table: **STUDENT_COURSE** (used by a university).

Roll_No	Student_Name	Dept	Dept_HOD	Course	Credits	Grade
101	Alice	CS	Dr. Smith	DBMS	4	A
102	Bob	CS	Dr. Smith	Python	4	B
103	Charlie	EC	Dr. Jones	DBMS	4	A

1. What is a Functional Dependency (FD)?

It is a **logical business rule**.

- **Rule:** If two rows have the same value on the Left Side (X), they **must** have the same value on the Right Side (Y).
- **Example FD:** `Roll_No → Student_Name`
- **Why?** Look at the table. If I give you Roll_No `101` , it always gives `Alice` . It is impossible for Roll_No 101 to be "Alice" in one row and "Bob" in another.

2. Trivial vs Non-Trivial FD

- **Trivial (Always true, useless):** `{Roll_No, Student_Name} → {Roll_No}`
- **Why?** The Right side (Roll_No) is just a part of the Left side. It is obvious. You don't need a rule for this.
- **Non-Trivial (Important business rule):** `{Roll_No} → {Student_Name}`
- **Why?** The Right side (Student_Name) is **not** a part of the Left side (Roll_No). This is a real rule that must be enforced.

3. Closure of Attributes (How to find the KEY)

Closure means: *"Starting with X, what is the FULL set of all columns I can reach?"*

- **Question:** Find the Closure of `Roll_No` . (Written as `Roll_No⁺`).
- **Step 1:** Start with itself. `{Roll_No}` .
- **Step 2:** Look at rules. `Roll_No → Student_Name` . Add it. Now we have `{Roll_No, Student_Name}` .
- **Step 3:** Look at more rules. `Roll_No → Dept` . Add it. Now we have `{Roll_No, Student_Name, Dept}` .
- **Step 4:** `Dept → Dept_HOD` . Add it. Now we have `{Roll_No, Student_Name, Dept, Dept_HOD}` .
- **Result:** `Roll_No⁺` = `{Roll_No, Student_Name, Dept, Dept_HOD}` . *(Note: It cannot find Course or Grade because one student takes many courses).*

4. Finding Candidate Key using Closure

A **Candidate Key** is the smallest set of columns whose Closure gives **ALL** columns of the table.

- **Try** `Roll_No` **:** `Roll_No⁺` gave `{Roll_No, Name, Dept, HOD}` .
  It **missed** `Course` , `Credits` , and `Grade` . So `Roll_No` alone is **NOT** a key.
- **Try** `Course` **:** `Course⁺` gives `{Course, Credits}` (because Course → Credits). It misses everything else. Not a key.
- **Try** `{Roll_No, Course}` **:**
  - `Roll_No` gives Name, Dept, HOD.
  - `Course` gives Credits.
  - Together, they give **ALL** columns: `{Roll_No, Name, Dept, HOD, Course, Credits, Grade}` .
- **Result:** `{Roll_No, Course}` is the **Candidate Key** (the unique identifier for this table).

5. Armstrong's Axioms (Math Rules for FDs)

If I tell you Rule 1: `X → Y` and Rule 2: `Y → Z` , you can mathematically guess Rule 3: `X → Z` without even looking at the table.

- **Reflexivity (Obvious):** If I know `{Roll_No, Course}` , I obviously know `{Roll_No}` . `{A,B} → {A}` .
- **Augmentation (Adding):** If `Roll_No → Name` , then adding `Course` to both sides still holds: `{Roll_No, Course} → {Name, Course}` .
- **Transitivity (Chain):** If `Roll_No → Dept` and `Dept → HOD` , then directly: `Roll_No → HOD` . (You don't need Dept in between).

6. Full vs Partial Dependency (Crucial for Normalization)

- **Full FD:** The Left side must have **ALL** the columns to find the Right side.
  - Example: `{Roll_No, Course} → Grade` . (You need BOTH student AND course to know their grade. One alone doesn't work). This is **Full**.
- **Partial FD:** A **subset** of the left side is enough to find the Right side.
  - Example: `{Roll_No, Course} → Student_Name` .
  - Why? Because a **subset** (`Roll_No`) alone is enough to find the name. You don't need the Course. This is **Partial**.

7. Transitive Dependency (Chain reaction)

When one non-key column depends on another non-key column (instead of directly on the key).

- We know: `Roll_No → Dept` and `Dept → HOD` .
- Therefore: `Roll_No → HOD` .
- **Problem:** `HOD` depends on `Dept` , not directly on the student. If Dr. Smith leaves, updating HOD in 1 place is fine, but in this table you'd have to update it in 1000 places! This is a **Transitive Dependency**.

Summary Cheat Sheet (Just for revision)

Topic	Explanation using our example
FD Rule	If you know `Roll_No` , you **must** know the `Name` .
Trivial	`Roll_No, Name → Name` (Useless, obvious).
Closure	`Roll_No⁺` = Roll_No + Name + Dept + HOD.
Candidate Key	`{Roll_No + Course}` because only together they find everything.
Partial Dep.	`{Roll_No + Course} → Name` (Roll_No alone finds Name). **BAD.**
Transitive Dep.	`Roll_No → Dept → HOD` . **BAD.**

Decomposition

1. What is Decomposition?

Splitting one big table into smaller tables to remove redundancy.

2. Two Must-Know Properties

Property	Meaning
Lossless Join	Joining sub-tables gives back **exact** original table (no extra rows)
Dependency Preservation	All original FDs are still checkable in sub-tables without joining

3. Lossless vs Lossy

3. Lossless vs Lossy

	Lossless (Good)	Lossy (Bad)
Join result	`R1 ⋈ R2 = R` (exact match)	`R1 ⋈ R2 ⊃ R` (extra fake rows)
Key test	Common attr must be **key** in at least one sub-table	Common attr is **not** a key in either

4. Dependency Preservation

- **Preserved:** All FDs can be checked locally in sub-tables.
- **Not Preserved:** Need to join tables to check some FDs.

5. Interview Quick Recap

Question	Answer
Which property is mandatory?	**Lossless Join**
What happens in lossy?	Extra rows appear on join
Lossless test?	Common column = key in at least one sub-table

Normalization

The Scenario: A University Course Enrollment Table

Our university has one giant table called **STUDENT_COURSE** that tracks students, their professors, and their grades.

Original Table R:

Student_ID	Student_Name	Course_ID	Course_Name	Professor_ID	Professor
101	Alice	C1	DBMS	P1	1111
101	Alice	C2	Python	P2	2222
102	Bob	C1	DBMS	P1	1111
103	Charlie	C3	Java	P3	3333

Functional Dependencies (Business Rules):

1. `Student_ID → Student_Name` (A student has one name)
2. `Course_ID → Course_Name, Professor_ID` (A course has one name and one professor)
3. `Professor_ID → Professor_Phone` (A professor has one phone)
4. `Student_ID, Course_ID → Grade` (A student gets one grade per course)

🔴 Step 0: The Problems (Why we need Normalization)

- **Redundancy:** Alice's name is repeated twice. Professor P1's phone is repeated.
- **Update Anomaly:** If Professor P1 changes phone to 9999, we must update it in 2 rows. If we miss one, data becomes inconsistent.
- **Insert Anomaly:** We cannot add a new course (`C4`) taught by `P4` until a student enrolls in it.
- **Delete Anomaly:** If Bob drops `C1` , we lose the fact that `P1` teaches `DBMS` .

🟢 Step 1: First Normal Form (1NF)

Rule: Every cell must have a **single atomic value** (no lists, no multiple values).

Our table is already in 1NF because every cell has exactly one value (e.g., Alice has two rows, not one row with two courses in one cell).

Status: ✅ **1NF achieved.**

🟡 Step 2: Second Normal Form (2NF)

Rules:

1. Must be in 1NF (✅ Yes).
2. **No Partial Dependencies.** (Meaning: No non-prime attribute should depend on only *part* of the composite key).

Find the Candidate Key:
`{Student_ID, Course_ID}` is the composite primary key (both needed to find the Grade).

Check for Partial Dependencies (The "Lazy Subset" test):

Attribute	Depends on?	Is it Partial?
`Student_Name`	Depends on `Student_ID` (a subset of the key)	✅ **YES (Partial)**
`Course_Name`	Depends on `Course_ID` (a subset of the key)	✅ **YES (Partial)**
`Professor_ID`	Depends on `Course_ID` (a subset of the key)	✅ **YES (Partial)**
`Professor_Phone`	Depends on `Professor_ID` (which depends on `Course_ID`)	✅ **YES (Transitive via Partial)**
`Grade`	Depends on **both** `Student_ID + Course_ID`	❌ **NO (Full)**

Fix (Decompose to remove Partial Dependencies):
Split the table into **three** tables:

- **R1 (Student_Info):** `Student_ID` , `Student_Name`
- **R2 (Course_Info):** `Course_ID` , `Course_Name` , `Professor_ID`
- **R3 (Enrollment):** `Student_ID` , `Course_ID` , `Grade`

Status: ✅ **2NF achieved.** (No partial dependencies remain).

🟠 Step 3: Third Normal Form (3NF)

Rules:

1. Must be in 2NF (✅ Yes).
2. **No Transitive Dependencies.** (Meaning: No non-prime attribute should depend on another non-prime attribute).

Check each table:

- **R1:** `Student_ID → Student_Name` . LHS is a super key. No transitive dependency. ✅ Safe.
- **R3:** `Student_ID, Course_ID → Grade` . LHS is a super key. No transitive dependency. ✅ Safe.
- **R2:** `Course_ID → Course_Name, Professor_ID` . LHS is a super key. ✅ Safe.
  - But wait! We have another FD: `Professor_ID → Professor_Phone` .
  - In R2, `Professor_ID` is **not** a super key. `Professor_Phone` is a non-prime attribute.
  - This means `Course_ID → Professor_ID → Professor_Phone` is a **Transitive Dependency**!

Fix (Decompose to remove Transitive Dependency):
Split **R2** into two tables:

- **R2a (Course_Info):** `Course_ID` , `Course_Name` , `Professor_ID`
- **R2b (Professor_Info):** `Professor_ID` , `Professor_Phone`

Status: ✅ **3NF achieved.** (No transitive dependencies remain).

🔵 Step 4: Boyce-Codd Normal Form (BCNF)

Rule:

1. Must be in 3NF (✅ Yes).
2. **For every non-trivial FD** `X → Y` **,** `X` **must be a super key.**

Check each table:

- **R1:** `Student_ID → Student_Name` . LHS (`Student_ID`) is a super key. ✅ Safe.
- **R2a:** `Course_ID → Course_Name, Professor_ID` . LHS (`Course_ID`) is a super key. ✅ Safe.
- **R2b:** `Professor_ID → Professor_Phone` . LHS (`Professor_ID`) is a super key. ✅ Safe.
- **R3:** `Student_ID, Course_ID → Grade` . LHS is a super key. ✅ Safe.

Status: ✅ **BCNF achieved.** (Every determinant is a super key).

Insert , Delete and Update **Anomalies** defination

- Unable to insert a new record into the table because **other required data is missing**.
- Deleting one record unintentionally **removes other valuable data** that you wanted to keep.
- Updating a value in **one row** requires updating it in **many rows** to keep data consistent. If you miss even one row, data becomes inconsistent.

Transactions

1. What is a Transaction?

Definition:
A transaction is a **single logical unit of work** that performs one or more operations on the database.

Example:
Transferring ₹500 from Account A to Account B is one transaction, but it has two operations:

1. Debit ₹500 from A (Write A)
2. Credit ₹500 to B (Write B)

Either **both** operations happen, or **neither** happens.

2. Operations in a Transaction

Operation	What it does	Where it works
Read(A)	Reads the value of `A` from the **database** and loads it into the **buffer** (main memory)	Memory
Write(A)	Writes the updated value of `A` from the **buffer** back to the **database**	Database

Important:
All changes made during a transaction are first stored in the **buffer (main memory)**. They only become permanent in the database when the transaction is **Committed**.

3. Transaction States (Life Cycle)

Here is the **complete flow** of a transaction from start to end:

```
Active → Partially Committed → Committed → Terminated
              ↓
         Failed → Aborted → Terminated
```

1. Active State

Definition:
The transaction is **currently executing** its instructions (Read/Write operations).
Where are changes stored? In the **buffer** (main memory), not yet in the database.

2. Partially Committed State

Definition:
The **last instruction** of the transaction has been executed, but the changes are **still in the buffer** (not yet written to the database).
Why "partially"? Because a failure can still occur before the changes are permanently saved.

3. Committed State

Definition:
All changes made by the transaction have been **successfully written** to the database.
Now: The transaction is **fully committed** and permanent. It cannot be undone.

4. Failed State

Definition:
A failure occurs during the **Active** or **Partially Committed** state, making it impossible to continue execution.
Causes: Hardware failure, power outage, deadlock, or logical errors.

5. Aborted State

Definition:
After failure, all changes made by the transaction must be **undone** (rolled back).
Process: The database is restored to its original state before the transaction started.
Result: The transaction is now aborted.

6. Terminated State

Definition:
The **final state** of any transaction.
How to reach:

- From **Committed** → Transaction completed successfully.
- From **Aborted** → Transaction failed and rolled back.

After this, the transaction's life cycle ends.

Here is your **complete, structured note** on **ACID Properties**, formatted exactly like the previous notes for easy revision.

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1. What are ACID Properties?

Definition:
ACID properties are a set of **rules** that ensure all transactions are processed reliably and the database remains **consistent** even in case of failures, concurrent access, or system crashes.

Acronym:

- **A**tomicity
- **C**onsistency
- **I**solation
- **D**urability

2. Atomicity (All or Nothing)

Definition:
A transaction must be executed **completely** or **not at all**. It cannot be left partially completed.

What it prevents:
If a failure occurs mid-transaction, all changes made so far are **rolled back** (undone). The database returns to its original state.

Example (Transfer ₹500 from A to B):

- Operations: Debit A (₹500) + Credit B (₹500)
- If credit to B fails after debiting A, the debit is **rolled back**. Both happen or neither happens.

Who handles it:
Transaction Management (using commit/rollback).

3. Consistency (Data Integrity)

Definition:
The database must remain in a **valid/consistent state** before and after the transaction. All integrity constraints (like primary key, foreign key, unique, check constraints) must be satisfied.

What it prevents:
Transactions cannot violate business rules or constraints.

Example (Account balance must never go negative):

- Before transaction: A = ₹500, B = ₹1000
- After transaction: A = ₹0, B = ₹1500 ✅ Consistent
- If A has ₹500 and you try to withdraw ₹600 → Transaction **aborts** because it violates the constraint.

Who handles it:
Application Developer (defines constraints) + **DBMS** (enforces them).

4. Isolation (Concurrency Control)

Definition:
Multiple transactions executing **simultaneously** should not interfere with each other. The final result should be the same as if they ran **one after another** (serially).

What it prevents:

- Lost updates
- Dirty reads (reading uncommitted data)
- Non-repeatable reads
- Phantom reads

Example (Two transactions at the same time):

- T1: Read A (₹500) → Add ₹100 → Write A (₹600)
- T2: Read A (₹500) → Add ₹200 → Write A (₹700)
- Without isolation, both read ₹500, and one update is **lost**.
- With isolation, one transaction waits until the other completes.

Who handles it:
Concurrency Control Manager (using locks, timestamps, or serializability).

5. Durability (Permanent Changes)

Definition:
Once a transaction is **committed**, all its changes become **permanent** and will survive any future system failures (power outage, crash, etc.).

What it ensures:
Committed data is safely stored on **disk** (not just in memory buffer). Even if the system crashes immediately after commit, the data remains.

Example (After successful transfer):

- Transaction commits: A = ₹0, B = ₹1500
- Immediately after commit, power goes out.
- When system restarts, A = ₹0 and B = ₹1500 are still there.

Who handles it:
Recovery Manager (using logs, write-ahead logging, and backups).

Q: Which ACID property is the most important?
A: All are mandatory, but **Atomicity** and **Durability** are considered the core guarantees of any DBMS.

Q: Can Isolation be relaxed?
A: Yes, for performance. But relaxed isolation levels (like Read Uncommitted) come with risks (dirty reads). It's a trade-off between performance and correctness.

Q: What happens if Isolation is not enforced?
A: Concurrency problems like dirty reads, lost updates, and inconsistent data.

Schedules & Serializability

1. Serial vs Non-Serial (The Basics)

- **Serial:** Transactions run one after another. Slow but **always safe**.
- **Non-Serial:** Transactions run together (interleaved). Fast but **can be unsafe**.

2. What is Serializable?

- A non-serial schedule is **safe** if it gives the same result as some serial schedule.
- **Golden Rule:** Serializable = Consistent.

3. Conflict Serializability (The #1 Interview Question)

- **How to check:** Two operations **conflict** if:
  1. Different transactions.
  2. Same data item.
  3. At least one is a **Write**.
- **Swap Rule:** If you can swap non-conflicting operations to make it serial → **Conflict Serializable**.
- **The Shortcut (Precedence Graph):**
  - Draw arrow `Ti → Tj` if `Ti` conflicts and comes before `Tj` .
  - **No cycle** = Conflict Serializable. ✅
  - **Cycle exists** = Not Conflict Serializable. ❌

4. Recoverability (Dirty Reads)

- **Dirty Read:** Reading data written by an uncommitted transaction.
- **Irrecoverable (BAD):** Transaction commits after reading dirty data, but the writer **aborts** later. Data is permanently wrong. ❌
- **Recoverable (GOOD):** Transaction **waits** to commit until the writer commits or aborts. ✅
- In an Irrecoverable schedule, a transaction reads a value written by an uncommitted transaction and commits before that transaction commits or aborts. If the writer later aborts, the reader has already committed with wrong data and cannot rollback. In a Recoverable schedule, the reader delays its commit until the writer either commits or aborts, ensuring no permanent inconsistency.

5. Cascading vs Cascadeless (The Follow-up)

- **Cascading (BAD):** One transaction fails → forces many others to rollback.
- **Cascadeless (GOOD):** You **cannot** read dirty data. You must wait for the writer to commit first.

Questions Interviewers Ask

Question	Your Answer
Is this schedule conflict serializable?	Draw precedence graph. If no cycl
Is this schedule recoverable?	Check if any transaction commits
What is the difference between Cascadeless and Strict?	Cascadeless = No dirty reads. Stric

Here is a **single, high-quality interview question** properly solved step-by-step. No extra theory, just pure solving.

The Question

Consider the following schedule S with two transactions T1 and T2:

Time	T1	T2
t1	Read(A)	
t2		Read(A)
t3	Write(A)	
t4		Write(A)
t5	Commit	
t6		Commit

Questions:

1. Is this schedule **Conflict Serializable**?
2. Is it **Recoverable**?

Solution

Step 1: Find Conflicting Operations

Rule: Conflict = Different transactions + Same data + At least one Write.

Pair	Operations	Conflict?
t1 (T1: Read A) & t2 (T2: Read A)	Read vs Read	❌ No
t1 (T1: Read A) & t4 (T2: Write A)	Read vs Write	✅ Yes (T1 before T2)
t2 (T2: Read A) & t3 (T1: Write A)	Read vs Write	✅ Yes (T2 before T1)
t3 (T1: Write A) & t4 (T2: Write A)	Write vs Write	✅ Yes (T1 before T2)

Step 2: Draw Precedence Graph

- From conflict 1: `T1 → T2`
- From conflict 2: `T2 → T1`
- From conflict 3: `T1 → T2`

Graph:

```
T1 ↔ T2 (Cycle: T1 → T2 → T1)
```

Step 3: Check for Cycle

- Yes, there is a **cycle** (`T1 → T2 → T1`).

Verdict 1:
❌ **Not Conflict Serializable.**
(It cannot be equivalent to any serial schedule.)

Step 4: Check Recoverability

Rule for Recoverable:
If a transaction reads dirty data (from an uncommitted transaction), it must **delay its commit** until the writer commits or aborts.

Check dirty reads:

- T2 reads A at **t2**.
- T1 writes A at **t3**.

Wait! T2 **read** before T1 wrote. So T2 **did not** read dirty data from T1.

Check the other way:

- T1 writes at **t3**.
- T2 writes at **t4**.
- T2 never reads what T1 wrote (T2 only writes).

No transaction reads uncommitted data.

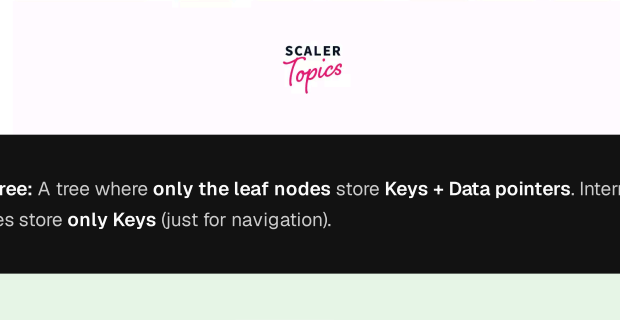
Verdict 2:
✅ **Recoverable.**
(No dirty reads happened, so no rollback is needed.)

Final Answers

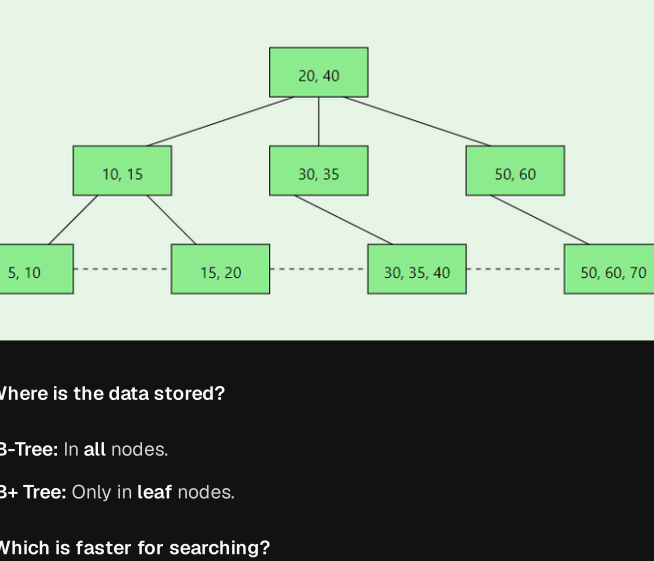
Question	Answer
Is it Conflict Serializable?	❌ **No** (Cycle in precedence graph)
Is it Recoverable?	✅ **Yes** (No dirty reads)

B- Tree and B+ Tree

- **B-Tree:** A tree where **every node** (root, middle, leaf) stores **Keys + Data pointers**.



- **B+ Tree:** A tree where **only the leaf nodes** store **Keys + Data pointers**. Internal nodes store **only Keys** (just for navigation).



Q1. Where is the data stored?

- **B-Tree:** In **all** nodes.
- **B+ Tree:** Only in **leaf** nodes.

Q2. Which is faster for searching?

- **B+ Tree.** Because internal nodes have no data, they can hold more keys → tree becomes shorter → fewer disk reads.

Q3. Which is better for range queries (e.g., `BETWEEN 10 AND 20` **)?**

- **B+ Tree.** Because leaf nodes are **linked** together. Just find the start and walk forward.

Q4. Why do databases use B+ Trees instead of B-Trees?

- **Faster searches** (shorter tree).
- **Efficient range queries** (linked leaves).
- **Predictable performance** (every search takes the same time).